CHETANKUMAR KHADKE

Staff Software Engineer - LLM, Model Fine-Tuning & Agentic AI

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PROFESSIONAL SUMMARY

Staff Software Engineer with 13 years in natural language processing and document intelligence, spanning custom model training and fine-tuning, transformer-based document models, and production LLM and agentic systems. At Xoriant, led the design, custom training, fine-tuning, and evaluation of the LayoutLM, Donut, UDOP, and Llama 2 models behind an enterprise Document AI platform, improving extraction accuracy by 70%. Now at Xpanse, custom-trains open-weight models for entity extraction and leads LLM and multi-agent integration on a distributed AWS pipeline, with context-aware guardrails, model security controls, and systematic model validation and verification across 500+ document types and millions of documents weekly, for a further ~30% accuracy gain. Hands-on with parameter-efficient fine-tuning (LoRA, QLoRA, PEFT), quantization, and accuracy benchmarking of open-weight models. Four peer-reviewed IEEE papers; certified in Machine Learning (Silver Medal) and Deep Learning by IIT Madras.

TECHNICAL SKILLS

Large Language Models (LLM) & Generative AI: Gemini 3 Pro, Gemini Flash, Claude Sonnet on Amazon Bedrock, GPT-OSS, Llama 2, Llama 3.2, Mistral AI, OpenAI, Qwen 2.5 VL, LLaVA, Llama-cpp, Hugging Face Transformers, LangChain, LlamaIndex, Prompt Engineering, RAG, Vision LLMs

Model Training & Fine-Tuning: Custom Model Training, Fine-Tuning, Parameter-Efficient Fine-Tuning (PEFT), LoRA, QLoRA, Instruction Tuning, Quantization (4-bit / 8-bit), Transfer Learning, Dataset Curation, Annotation & Labeling Workflows, Hyperparameter Tuning, GPU Training, PyTorch, Scikit-learn, Hugging Face Transformers & PEFT, MLflow

Model Guardrails, Security & Validation: Context-Aware Guardrails, Prompt Injection & Jailbreak Mitigation, Hallucination Mitigation, PII Redaction, Content Filtering, Model Validation & Verification, Model Governance, Ground-Truth Benchmarking, Accuracy / Precision / Recall / F1, Confidence Scoring, Human-in-the-Loop Review, Drift Monitoring

Agentic AI: Multi-Agent Systems, Agent Orchestration, LangGraph, Stateful Agent Graphs, Tool/Function Calling, Model Context Protocol (MCP), Agentic RAG, Human-in-the-Loop, Agent-to-Agent (A2A) Communication, Planning & Reasoning, Memory Management

NLP & Document AI: LayoutLM, Donut, UDOP, BERT, DocVQA, Named Entity Recognition (NER), Named Entity Linking (NEL), Text Classification, Clustering, Topic Modeling, Sentiment Analysis, Embeddings, Vector Search, Semantic Similarity, spaCy, NLTK, Sentence-Transformers, BERTopic, FastText

MLOps & Cloud: AWS (SageMaker, Bedrock, Bedrock AgentCore, AgentCore Runtime, Strands Agents SDK, ECR, ECS, S3, CloudWatch, Inference Endpoints), GCP, Docker, MLflow, Git, CI/CD, Model Deployment Pipelines, Containerization, Model Monitoring & Observability, REST APIs, Distributed Systems

Data Engineering & Databases: SQL, MySQL, MongoDB, Neo4j, Elasticsearch, Knowledge Graphs, Data Pipelines

Programming Languages: Python, R, Java

PROFESSIONAL EXPERIENCE

Staff Software Engineer — Feb 2025 - Present

Xpanse — *New Jersey, USA*

Cloud-based distributed Document AI platform on AWS (S3, SQS, ECS Fargate) for automated classification and data extraction across 500+ document types and 500+ data fields, processing millions of documents weekly.

- Led LLM integration for the production extraction pipeline, raising end-to-end field extraction accuracy by approximately 30% across 500+ field types, while mentoring engineers on LLM evaluation and prompt design.
- Custom-trained an open-weight GPT-OSS model for entity extraction, applying parameter-efficient fine-tuning (LoRA/QLoRA) over curated in-domain ground truth to raise field-level accuracy and cut inference cost.
- Designed and shipped a production multi-agent orchestration layer on LangGraph, coordinating classification, extraction, and validation agents through tool/function calling and Model Context Protocol (MCP) integrations.
- Engineered and fine-tuned prompt libraries for Gemini 3 Pro, Gemini Flash, and Claude Sonnet on Amazon Bedrock, delivering precise, context-rich extraction across heterogeneous document layouts.
- Introduced context-aware guardrails and model security controls - prompt-injection defenses, PII handling, and policy checks - cutting hallucination rates and producing auditable outputs on business-critical fields.

- Established a model validation, verification, and governance framework: ground-truth benchmark sets, per-field accuracy and precision tracking, and confidence-based routing to human review, gating every model and prompt change.
- Built a framework that generates field-specific prompts on demand from document-field mappings, significantly cutting onboarding time for new document types.

Xoriant Solutions
May 2021 - Feb 2025

Software Engineer
Aug 2024 - Feb 2025 | New Jersey, USA

Trade document extraction and reconciliation across contracts, invoices, bills of lading, and financial statements, with real-time discrepancy detection.

- Created and tuned context-aware prompt strategies on Gemini 1.5 Pro for high-precision extraction from unstructured trade and financial documents, mitigating hallucination risk on data where errors carry direct monetary cost.
- Designed reconciliation status checks combining LLM reasoning with sentence-embedding similarity matching to detect and report cross-source discrepancies in real time.

Technology Specialist
May 2021 - Jul 2024 | Pune, India

Built the core document-understanding model layer for an enterprise Document AI platform.

- Led the design, custom training, fine-tuning, and evaluation of document-understanding models on LayoutLM, Donut, UDOP, and Donut-based DocVQA, achieving a 70% improvement in data extraction accuracy.
- Trained the LLaMa-v2 based model for extraction and InternVL based model for classification.
- Spearheaded a Named Entity Recognition module, benchmarking NER libraries against custom-trained models on held-out ground truth, and established the annotation and labeling workflows behind the training data.
- Custom-trained a Llama 2 model for entity extraction on domain-specific documents, using instruction tuning and parameter-efficient fine-tuning to generalize to unseen and low-template layouts.
- Owned model training and deployment pipelines end to end, tracking experiments and accuracy metrics in MLflow and containerizing models with Docker for CI/CD release to AWS ECR/ECS.

Engineering Lead
Aug 2018 - May 2021

Persistent Systems
Pune, India

Knowledge graph construction and conversational AI for enterprise data discovery.

- Led development of a knowledge graph linking organizations, individuals, and content across diverse data sources, modeled and served on Neo4j to enable relationship discovery.
- Built custom NER and relation-extraction classification models, and performed Named Entity Linking against Wikipedia using spaCy to populate and disambiguate graph entities.
- Developed the intent-classification engine for an automated customer-support chatbot using Scikit-learn SVM, Naive Bayes, and RNN models combined with sentence-similarity features.

Systems Engineer to IT Analyst
NOV 2013 - Aug 2018

Tata Consultancy Services
Pune, India

Applied NLP/ML research and delivery for the ignio cognitive automation product and TCS Research.

- Built a system that mines process and workflow models from system integration test cases, using NLP preprocessing and ML classification to recover software behavior from test artifacts.

EDUCATION

Bachelor of Technology, Computer Science 2009 - 2013
Dharmsinh Desai University, Nadiad, Gujarat, India

CERTIFICATIONS & AWARDS

- Machine Learning (Silver Medal) and Deep Learning - Indian Institute of Technology Madras (NPTEL)
- Bravo Award, received 3 times for outstanding work - Persistent Systems
- XOR-Champ Award for outstanding work - Xoriant Solutions

RESEARCH PUBLICATIONS

1. Khadke C., Rana S., Shah V. "Recovering Workflows from Functional Tests" - IEEE ICPC, Florence, Italy, 2015.

2. Shah V., Khadke C., Rana S. "Mining Process Models and Architectural Components from Test Cases" - IEEE ICSTW, Graz, Austria, 2015.

3. Shah V., Khadke C., Rana S. "Knowledge Transition: Discovering Models from Functional Tests" - IEEE ISSRE, Gaithersburg, MD, USA, 2015.

4. Khadke C. "Knowledge Acquisition (KnowX)" tool demonstration - ENASE, Rome, Italy, 2016.

Full publication list: dblp.uni-trier.de/pers/hd/k/Khadke:Chetan
Technical writing: medium.com/@khadkechetan